

Q1: For the following lines of code

```
sub  x1,  x4,  x5
add  x2,  x6,  x7
or   x3,  x1,  x2
```

- 1a. EX/MEM.RegisterRd = ID/EX.RegisterRs1
- 1b. EX/MEM.RegisterRd = ID/EX.RegisterRs2
- 2a. MEM/WB.RegisterRd = ID/EX.RegisterRs1
- 2b. MEM/WB.RegisterRd = ID/EX.RegisterRs2

1. Draw a multi-cycle clock diagram and identify type of data hazards for *add* and *or* instructions.

Sol:

add → No hazard
or → x1 has 2a whereas x2 has 1b

2. Re-write the code after inserting nops such that we don't have any hazards.

Sol:

```
sub x1, x2, x5
add x2, x6, x7
nop
nop
or x3, x1, x2
```

3. For *or* instruction, write control values for ForwardA and ForwardB multiplexer signals. In which clock cycle, these values will be used?

Instr	0	1	2	3	4	5	6
sub	F	D	E	M	W		
add		F	D	E	M	W	
or			F	D	E	M	W
ForwardA	...	...	0 0	0 0	0 1	...	...
ForwardB	...	...	0 0	0 0	1 0	...	...

Mux control	Source	Explanation
ForwardA = 00	ID/EX	The first ALU operand comes from the register file.
ForwardA = 10	EX/MEM	The first ALU operand is forwarded from the prior ALU result.
ForwardA = 01	MEM/WB	The first ALU operand is forwarded from data memory or an earlier ALU result.
ForwardB = 00	ID/EX	The second ALU operand comes from the register file.
ForwardB = 10	EX/MEM	The second ALU operand is forwarded from the prior ALU result.
ForwardB = 01	MEM/WB	The second ALU operand is forwarded from data memory or an earlier ALU result.